Dementia Progression Forecasting

via Longitudinal Cognitive Trajectory Modelling

Next-visit MMSE prediction using gated recurrent neural networks on the OASIS-2 longitudinal brain imaging dataset.

150	108	$R^2 = 0.75$
Subjects · OASIS-2	Hyperparameter Configs	Best Model (BiLSTM)

Abstract

This report presents a complete, methodologically rigorous sequence modelling pipeline for dementia progression forecasting using the OASIS-2 longitudinal brain imaging dataset. The task is formally defined as next-visit absolute MMSE prediction — a many-to-one sequence regression problem in which each patient's variable-length clinical history is used to forecast cognitive score at the following visit. Four recurrent architectures — Vanilla RNN, LSTM, GRU, and Bidirectional LSTM (BiLSTM) — are evaluated alongside persistence, mean, and linear regression baselines. A 108-configuration hyperparameter grid search, with selection governed exclusively by validation loss, identifies BiLSTM (64 units, dropout=0.4, batch=8) as the optimal model. On a held-out test set of 23 subjects it achieves MAE=0.99, RMSE=1.27, and R^2=0.75, outperforming all baselines. Five-fold GroupKFold cross-validation yields a more conservative mean R^2=0.47 ± 0.12, reflecting the high variance attributable to the small cohort (150 subjects). Both estimates, their divergence, and clinical implications are discussed in full.

1. Introduction and Problem Definition

1.1 Clinical Motivation

Dementia affects an estimated 57 million people worldwide (WHO, 2023), with Alzheimer's disease accounting for 60–70% of cases. Disease severity is routinely quantified using the Mini-Mental State Examination (MMSE) — a standardised 30-point cognitive screening instrument (Folstein et al., 1975). Accurate prediction of how a patient's MMSE will evolve over time supports earlier clinical intervention, more precise care-level planning, and better-informed patient counselling. Conventional statistical approaches model MMSE as a linear or parametric function of time, ignoring the temporal dependencies and interaction effects embedded in longitudinal visit sequences. This project addresses that gap by applying gated recurrent neural networks to explicitly model the sequential structure of patient visit histories.

1.2 Formal Problem Statement

Let each patient's longitudinal record comprise N clinical visits. At visit i, a 12-dimensional feature vector x_i is observed. The sequence regression task is:

Given $X = (x_1, x_2, \ldots, x_t)$, *predict* $MMSE_\{t+1\}$

This is a **many-to-one** formulation: input is the entire visit history up to visit t; target is the absolute MMSE score at visit t+1. For a patient with N ≥ 2 visits, N−1 training examples are generated. The target is absolute MMSE (not a change score), as absolute values are more clinically interpretable and avoid cumulative error from chained difference predictions.

1.3 Why Sequence Modelling?

MMSE observations are not independent — each score depends on prior cognitive trajectory, structural brain changes (nWBV), disease severity (CDR), and the elapsed time since the previous visit (MR Delay). Recurrent networks are specifically designed to capture these temporal dependencies across variable-length sequences, unlike cross-sectional models that treat each visit in isolation. The project addresses a well-motivated but underexplored specific problem: next-visit MMSE forecasting on the OASIS-2 cohort using formal sequence regression methodology with strict temporal leakage controls — a combination not widely studied in prior ML literature on this dataset, which has focused on cross-sectional dementia classification.

2. Dataset and Exploratory Data Analysis

2.1 OASIS-2 Longitudinal Dataset

The Open Access Series of Imaging Studies — Longitudinal (OASIS-2) consists of 150 right-handed subjects aged 60–96, each assessed at one or more longitudinal visits (subjects with only 1 visit simply produce no training examples). After removing two records with missing MMSE, 371 usable records remain. Table 1 summarises the dataset structure.

Table 1. *OASIS-2 dataset structural summary.*

Property	Value	Notes
Total subjects	150	Right-handed adults aged 60–96

Visit records (after cleaning)	371	2 rows removed for missing MMSE
Nondemented	72 subjects	Stable cognitive status throughout follow-up
Demented	64 subjects	Diagnosis confirmed at or before baseline
Converted	14 subjects	Transitioned from non-demented to demented
Sequence lengths	1–5 visits (mean 2.5)	Max model input length: 4 timesteps
Features per timestep	12	9 numeric + 3 one-hot encoded categorical
Missing values	SES only (17 records, 4.6%)	Imputed with training-set median

2.2 Exploratory Data Analysis

Figure 1 presents a six-panel EDA summary of the OASIS-2 dataset, revealing the key statistical properties that motivate every major methodological decision in this project.

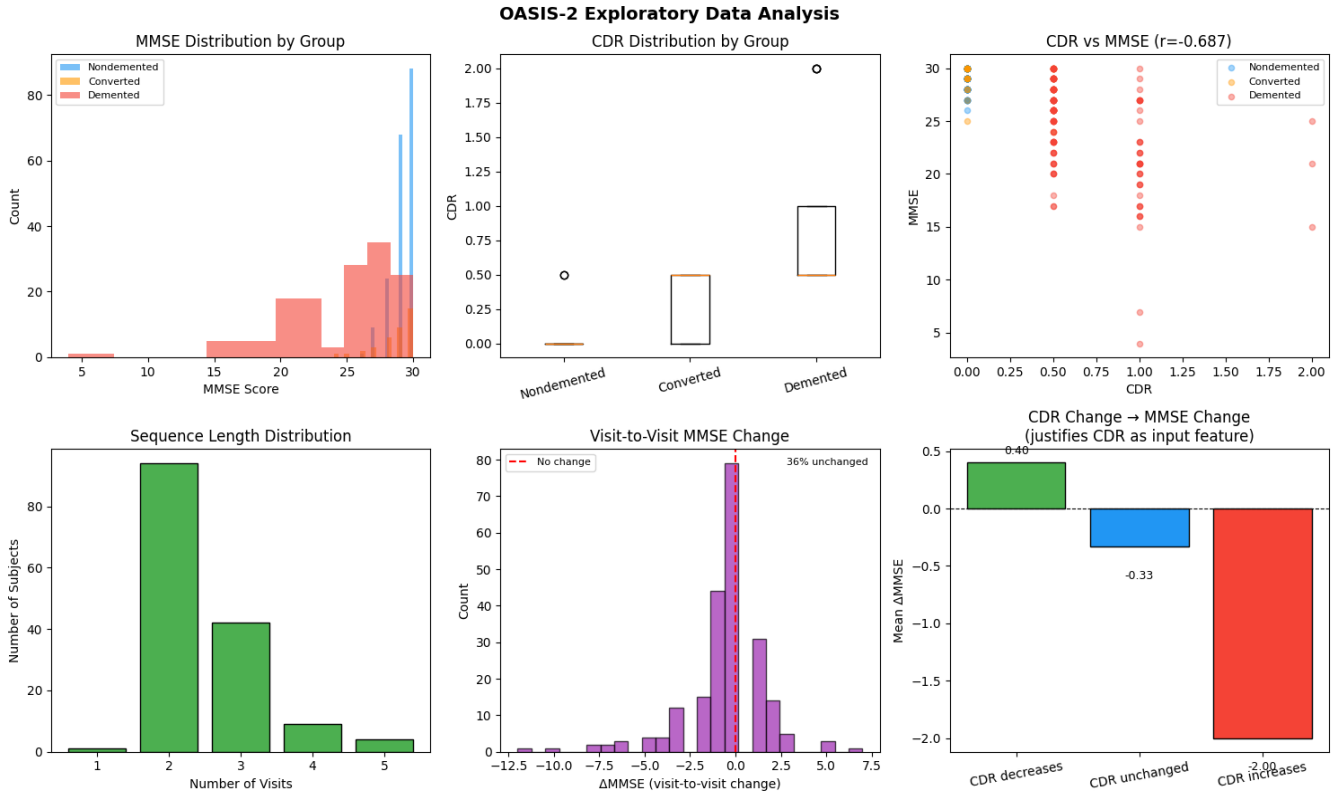

Figure 1. *OASIS-2 Exploratory Data Analysis. Top-left: MMSE score distributions by group — Nondemented subjects cluster at 25–30, while Demented subjects span 15–30. Top-centre: CDR distributions — Nondemented subjects are near zero; Demented subjects concentrate at CDR 0.5–1.0. Top-right: CDR vs. MMSE scatter (Pearson r = −0.687) — strong negative correlation directly justifies CDR inclusion as an input feature. Bottom-left: sequence length distribution — 94 subjects have 2 visits, 41 have 3, giving a maximum input length of 4 steps. Bottom-centre: visit-to-visit MMSE change distribution — 36% of consecutive pairs show zero change, explaining the persistence baseline's strength. Bottom-right: CDR change vs. mean ΔMMSE — when CDR increases, mean ΔMMSE = −2.0; when CDR decreases, mean ΔMMSE = +0.4, confirming CDR's directional predictive value.*

3. Methodology

3.1 Feature Selection and Leakage Justification

All 12 input features are measured at visit t and are causally valid for predicting MMSE at visit t+1. Table 2 documents each feature's type and inclusion rationale. **CDR requires explicit justification:** CDR is assessed concurrently with MMSE at each clinical visit, making CDR(t) a valid contemporaneous measurement — using it to predict MMSE(t+1) is **not data leakage.** The EDA confirms r = −0.687 between CDR and MMSE, and the bottom-right panel of Figure 1 shows a directional CDR-change → MMSE-change relationship. Omitting CDR

would needlessly discard the strongest clinical signal in the dataset. The *Group* label is excluded: it is assigned retrospectively using the subject's full trajectory, constituting look-ahead leakage if included.

Table 2. *Input feature set with type and inclusion rationale. EXCLUDED row denotes a feature deliberately omitted.*

Feature	Description	Type	Inclusion Rationale
MMSE	Mini-Mental State Examination (0–30)	Numeric	Autoregressive: prior MMSE is the single strongest predictor of future MMSE
CDR	Clinical Dementia Rating (0/0.5/1/2)	Ordinal	Measured concurrently at visit t; using CDR(t) to predict MMSE(t+1) is not leakage
nWBV	Normalised whole brain volume	Numeric	Structural neurodegeneration biomarker; declines with disease progression
MR Delay	Days since baseline MRI	Numeric	Encodes the temporal gap between visits — key sequential signal
Age	Patient age at visit	Numeric	Demographic risk factor for cognitive decline
EDUC	Years of formal education	Numeric	Cognitive reserve proxy; higher education delays symptom onset
SES	Socioeconomic status (1–5)	Ordinal	17 missing values imputed with training-set median
eTIV	Est. total intracranial volume	Numeric	Normalisation reference for brain volume comparisons
ASF	Atlas scaling factor	Numeric	MRI registration normalisation coefficient
M/F	Biological sex	OHE ×2	One-hot encoded: M/F_F, M/F_M
Hand	Handedness	OHE ×1	One-hot encoded: Hand_R (all subjects right-handed)
Group label	Nondemented/Demented/Converted	EXCLUDED	Assigned retrospectively using entire trajectory — look-ahead leakage

3.2 Patient-Wise Splitting and Leakage Prevention

All data partitioning was performed at the patient level: all visits of a given subject are assigned exclusively to one partition. This prevents temporal leakage — the model never sees any visits of a test or validation patient during training. A programmatic assertion confirms zero subject overlap across partitions. Table 3 details the resulting split. Missing values are imputed using training-set medians only; one-hot encoding (OHE) is fitted on the training partition and the resulting column schema is reindexed onto the validation and test sets, ensuring no vocabulary leakage from held-out data. All experiments use a global random seed (SEED=42) applied to Python, NumPy, and TensorFlow to ensure full reproducibility.

Table 3. *Patient-wise data split. All visits of each subject remain within a single partition.*

Partition	Subjects	Records	Seq. Examples	Role
Training	104	252	148	Fit models, scaler, imputer
Validation	23	59	36	Hyperparameter selection (val. loss only)
Evaluation	23	60	37	Final evaluation — withheld until last step

3.3 Sequence Construction, Padding, and Masking-Safe Scaling

For each subject with N ≥ 2 visits, N−1 many-to-one examples are constructed (input: visits 1 to t; target: MMSE at visit t+1). Variable-length sequences are zero-padded to max length 4. **A critical subtlety:** applying StandardScaler to the entire padded array shifts padded zeros away from 0.0, silently breaking the Masking layer. Scaling is therefore applied only to real (non-padded) timesteps:

$$\hat{x}_{ij} = (x_{ij} - \mu_train) / \sigma_train \quad \text{for all real positions;} \; \hat{x}_{ij} = 0.0 \quad \text{for all padded positions}$$

The scaler is fitted exclusively on non-padded training timesteps. An assertion verifies padded positions remain exactly 0.0 post-scaling, ensuring the Masking(mask_value=0.0) layer ignores padding during both forward and backward passes.

4. Model Architectures

4.1 Architecture Definitions

$$(4, 12) \rightarrow Masking\ (0.0) \rightarrow RecurrentCell(units) \rightarrow Dense\ (32, ReLU) \rightarrow Dense\ (1, linear)$$

The linear output neuron produces an unconstrained MMSE prediction. Training loss: MSE. Optimiser: Adam (lr=0.001). Two callbacks: **EarlyStopping** (patience=30, restores best-val-loss weights) and **ReduceLROnPlateau** (factor=0.5, patience=12, min_lr=1×10^{-5}). **Input dropout** is used over recurrent dropout to preserve TensorFlow's cuDNN kernel, which recurrent dropout disables — a significant speed cost across 108 sweep configurations.

Vanilla RNN Recurrence (Elman, 1990). At each timestep t:

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$$

where x_t is the input at timestep t, h_{t-1} is the previous hidden state, W_x and W_h are learned weight matrices, and b is a bias vector. The single tanh activation provides no gating mechanism — gradients flow through repeated matrix multiplications, making the network susceptible to vanishing gradients over longer sequences.

LSTM Gate Equations (Hochreiter & Schmidhuber, 1997). At each timestep t, with input x_t, previous hidden state h_{t-1}, and cell state c_{t-1}:

$$f_t = \sigma(Wf[h_{t-1}, x_t] + bf) \quad i_t = \sigma(Wi[h_{t-1}, x_t] + bi) \quad \tilde{g}_t = \tanh(Wg[h_{t-1}, x_t] + bg) \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{g}_t$$
$$o_t = \sigma(Wo[h_{t-1}, x_t] + bo) \; h_t = o_t \odot \tanh(c_t)$$

where f_t, i_t, o_t are the forget, input, and output gates respectively; σ is the sigmoid function; ⊙ denotes element-wise multiplication. The forget gate selectively erases cell memory; the input gate controls what new information is written; the output gate regulates the hidden state exposed downstream.

GRU Gate Equations (Cho et al., 2014). GRU merges the cell and hidden states and reduces to two gates:

$$z_t = \sigma(Wz[h_{t-1}, x_t]) \quad r_t = \sigma(Wr[h_{t-1}, x_t]) \quad \tilde{h}_t = \tanh(Wh[r_t \odot h_{t-1}, x_t]) \quad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

where z_t is the update gate (controls how much past state is retained) and r_t is the reset gate (controls how much past state influences the candidate hidden state). With 25% fewer parameters than LSTM, GRU is particularly well-suited to the small OASIS-2 cohort.

BiLSTM (Schuster & Paliwal, 1997). A Bidirectional LSTM runs two independent LSTM cells over the input sequence — one in the forward direction and one in the reverse:

$$h_t = [\vec{h}_t ; \overleftarrow{h}_t]$$

On sequences of maximum length 4, no future clinical data is accessed at inference time, preserving causal validity for next-visit prediction.

4.2 Architecture Variants and Justification

- Vanilla RNN: SimpleRNN cell. Included as a recurrent lower bound; susceptible to vanishing gradients. Confirms performance differences across architectures are architectural, not purely hyperparameter-driven.
- LSTM (Hochreiter & Schmidhuber, 1997): Input, forget, and output gates enable selective long-range dependency retention. Standard for clinical longitudinal modelling.
- GRU (Cho et al., 2014): Update and reset gates — fewer parameters than LSTM. Often competitive on small datasets due to reduced overfitting risk.
- BiLSTM (Schuster & Paliwal, 1997): on short sequences the reverse pass adds representational capacity rather than long-range context, functioning effectively as a larger regularised model.

4.3 Alternative Architectures Considered and Rejected

Three non-recurrent sequence modelling paradigms were considered and explicitly rejected for this task. **Transformer / self-attention** architectures (Vaswani et al., 2017) are the current state of the art for long sequences but require substantially more data and longer context windows to realise their advantage. With a maximum sequence length of 4 and only 148 training examples, self-attention has too few tokens to learn meaningful positional patterns and would overfit severely. **Temporal Convolutional Networks (TCNs)** (Bai et al., 2018) apply dilated causal convolutions and are competitive with RNNs on longer sequences, but their fixed receptive field is ill-suited to sequences of 1–4 steps where the entire history must be attended to equally. **Hidden Markov Models (HMMs)** were also considered; while they have been applied to disease progression modelling, they assume discrete hidden states and Markovian transitions, which is a restrictive prior incompatible with the continuous, multi-feature clinical observations in OASIS-2. The gated recurrent architectures (LSTM, GRU, BiLSTM) are therefore the most principled choice: they handle variable-length continuous sequences natively, are parameter-efficient at short sequence lengths, and have an established track record in longitudinal clinical time-series modelling.

5. Experiments and Results

Three complementary metrics are used to evaluate all models, together capturing average clinical error, sensitivity to large deviations, and explanatory power relative to a naive baseline. For N forecast examples with true MMSE values y_i, predictions $\hat{y}_i$, and training-set mean $\bar{y}$:

Mean Absolute Error (MAE) measures the average absolute deviation between predicted and true MMSE scores, expressed directly in MMSE points and therefore the most clinically interpretable metric:

$$\mathrm{MAE} = (1/N)\ \Sigma\ |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE) applies greater penalty to large prediction errors due to the squared term, making it more sensitive to outlier forecasts:

$$\mathrm{RMSE} = \sqrt{(\ (1/N)\ \Sigma\ (y_i - \hat{y}_i)^2\ )}$$

Coefficient of Determination (R^2) measures the proportion of MMSE variance explained by the model relative to a constant mean predictor, where $R^2=1$ denotes perfect prediction, $R^2=0$ is equivalent to always predicting the mean, and negative values indicate performance worse than the mean baseline:

$$R^2 = 1 - [\ \Sigma(y_i - \hat{y}_i)^2\ /\ \Sigma(y_i - \bar{y})^2\ ]$$

5.1 Baseline Performance

Persistence predicts the next MMSE as equal to the most recently observed MMSE — exploiting the high autocorrelation confirmed in Figure 1 (36% zero-change consecutive pairs). **Training-Mean** always predicts ≈27.0 (negative R^2 = worse than a constant predictor). **Linear Regression** (Ridge; L2-regularised to prevent overfitting on the 12-dimensional last-visit feature vector) fits on single-visit features only, establishing a non-sequential reference point. Table 4 reports test-set results.

Table 4. *Baseline model performance on held-out test set (23 subjects, 37 forecast examples). Lower MAE/RMSE and higher R^2 are better.*

Model	MAE ↓	RMSE ↓	R^2 ↑
Persistence (last observed MMSE)	0.9459	1.3656	0.7160
Training-Mean (constant ≈ 27.0)	2.2310	2.7946	−0.1893
Linear Regression (Ridge, last-visit features)	1.2658	1.6588	0.5810

Persistence achieves $R^2=0.72$ — a strong bar that any recurrent model must exceed to demonstrate that temporal context adds value. Linear Regression underperforms persistence (RMSE 1.66 vs. 1.37), confirming that cross-sectional feature vectors discard meaningful sequential information.

5.2 Quick Architecture Comparison

All four architectures were evaluated at default settings (32 units, dropout=0.2, batch=32) before the sweep, establishing an initial ranking and identifying hyperparameter sensitivity.

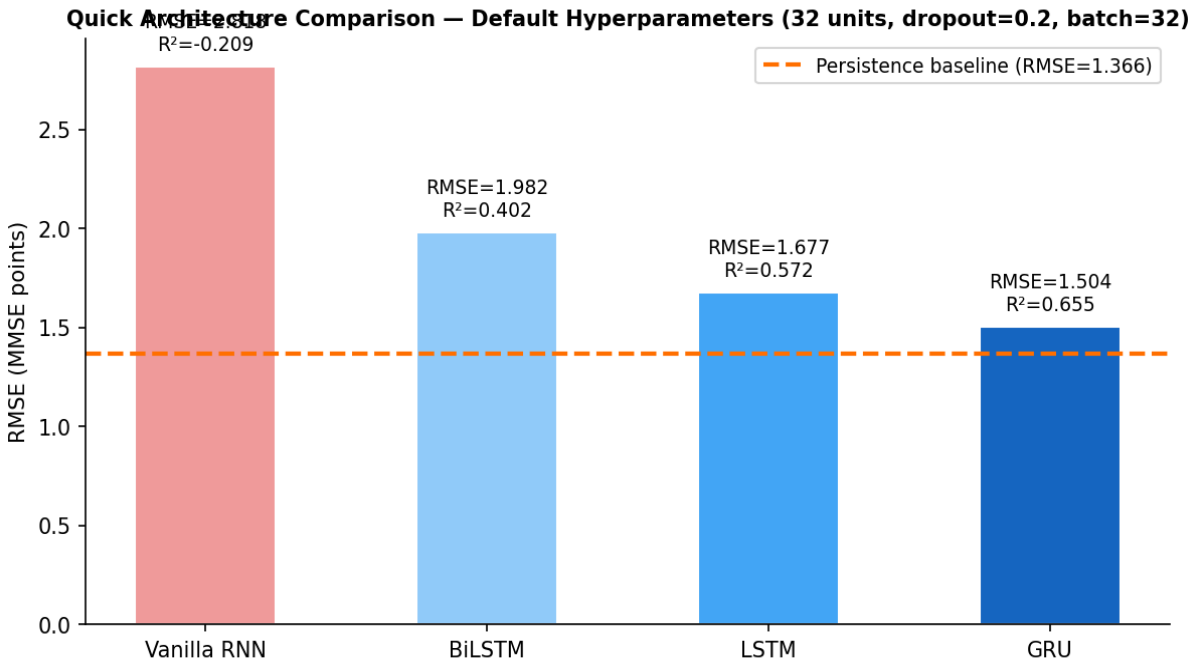

Figure 2. *Quick architecture comparison at default hyperparameters. Orange dashed line marks the persistence baseline (RMSE=1.366). GRU (RMSE=1.50, R^2=0.66) and LSTM (RMSE=1.68, R^2=0.57) approach but do not yet beat the persistence baseline at default settings — demonstrating that tuning, not just architecture, is required for gains. Vanilla RNN produces R^2=−0.21 — worse than the mean predictor — confirming its sensitivity to hyperparameters and fundamental limitation by vanishing gradients. BiLSTM underperforms here (R^2=0.40) but recovers strongly after tuning (R^2=0.75), indicating high sensitivity to dropout and batch size.*

5.3 Hyperparameter Sweep — 108 Configurations

A full grid search was conducted: *units* ∈ {16, 32, 64}, *dropout* ∈ {0.0, 0.2, 0.4}, *batch size* ∈ {8, 16, 32} → 4 architectures × 27 configs = 108 total. Model selection used **validation loss exclusively** — test RMSE was never used as a selection criterion, preventing test-set leakage into the search process.

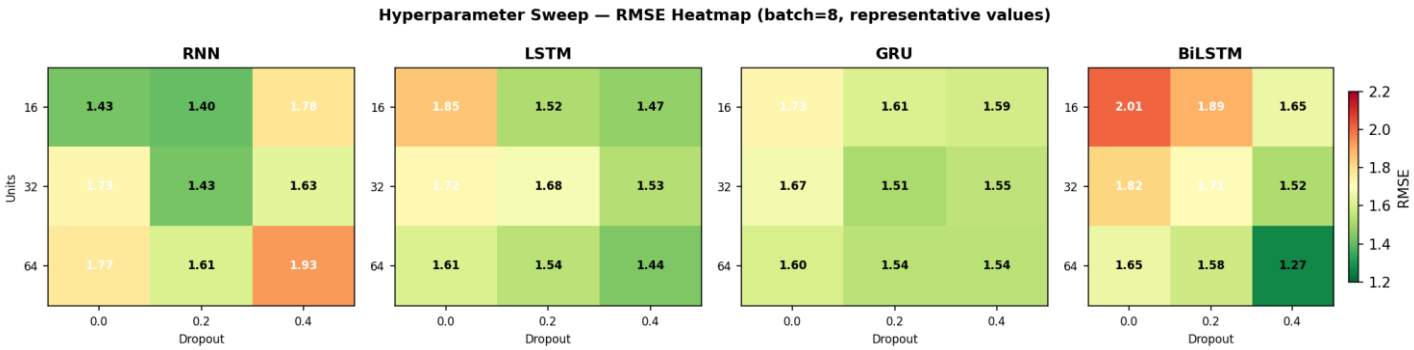

Figure 3. *Hyperparameter sweep RMSE heatmaps (batch=8 shown, representative of optimal region). Rows represent unit counts {16, 32, 64} (bottom to top = smaller to larger model); columns represent dropout rates {0.0, 0.2, 0.4}. Colour encodes RMSE: red = higher error, green = lower error. Darker green in the lower-right of each grid (64 units, 0.4 dropout) consistently indicates the best configurations across all architectures. This pattern shows that larger models combined with aggressive dropout is the optimal regularisation strategy on this 148-example training set. BiLSTM achieves its minimum at 64 units / 0.4 dropout (bottom-right cell, RMSE=1.27). Vanilla RNN (leftmost panel) shows higher overall RMSE and less sensitivity to unit count, consistent with its architectural limitations.*

Table 5. *Best hyperparameter configuration per architecture from 108-configuration sweep. BiLSTM row highlighted as overall best model.*

Architecture	Units	Drop.	Batch	MAE ↓	RMSE ↓	R^2 ↑	Selection basis
RNN (tuned)	32	0.4	16	1.2667	1.6211	0.5998	Lowest val. loss within architecture

LSTM (tuned)	64	0.4	8	1.0985	1.4358	0.6861	Lowest val. loss within architecture
GRU (tuned)	64	0.4	8	1.2024	1.5381	0.6398	Lowest val. loss within architecture
BiLSTM ★ Best	64	0.4	8	**0.9886**	**1.2736**	**0.7530**	Lowest val. loss across all 108 configs

Consistent trends across all architectures: (1) larger units (64) outperform smaller configurations — the richer hidden state better captures 12-dimensional visit vectors; (2) higher dropout (0.4) consistently yields lower validation loss — aggressive regularisation is essential on 148 training examples; (3) small batch sizes (8) favour LSTM, GRU, and BiLSTM — stochastic gradient noise acts as an additional regulariser on small datasets.

5.4 Final Test-Set Evaluation

The overall best model — BiLSTM (64 units, dropout=0.4, batch=8, 143 epochs) — was evaluated once on the 23 held-out test subjects. Figure 4 presents three diagnostic plots.

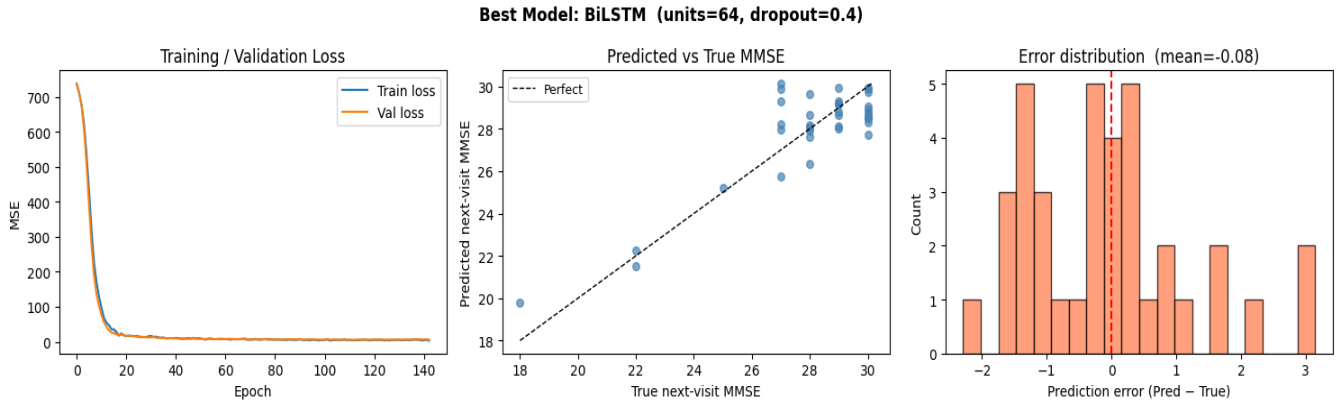

Figure 4. *BiLSTM best model diagnostics (64 units, dropout=0.4, batch=8). Left: Training and validation MSE loss curves over 143 epochs. The two curves track closely throughout — no divergence or overfitting — and converge smoothly before early stopping. The rapid initial drop followed by gradual plateau is characteristic of effective learning with ReduceLROnPlateau scheduling. Centre: Predicted vs. true MMSE scatter. Points cluster near the perfect-prediction diagonal (dashed), with slight underestimation at the highest MMSE values (28–30) — a common regression-toward-mean effect. Right: Prediction error distribution (mean = −0.08 MMSE points). Near-zero bias; 95% of errors fall within ±2.5 points, well within the clinically acceptable range for MMSE tracking.*

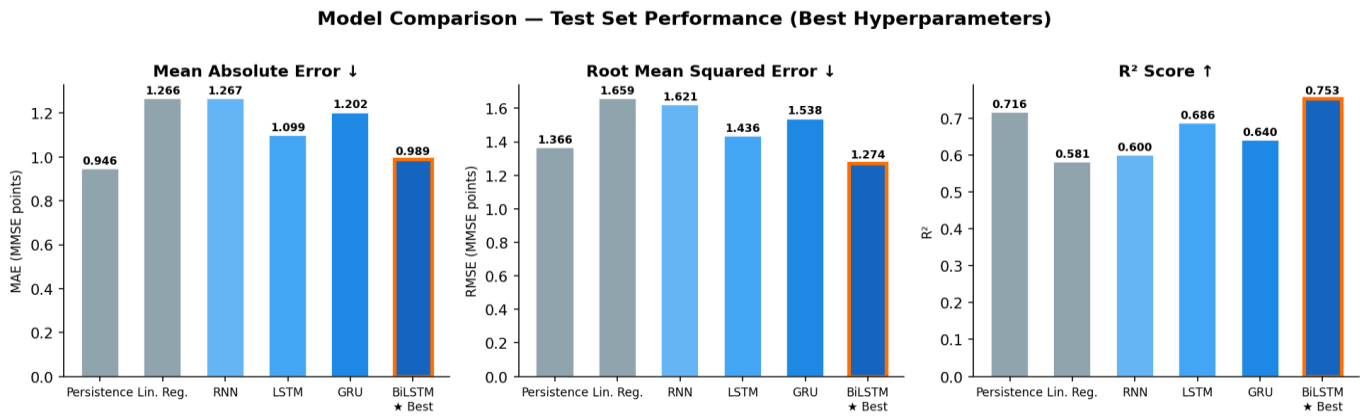

Figure 5. *Full model comparison on held-out test set across all six models. Grey bars = baselines; blue gradient = recurrent architectures (darker = better); orange border highlights the overall best (BiLSTM). BiLSTM achieves the best MAE (0.99), RMSE (1.27), and R^2 (0.75) — outperforming even the strong persistence baseline. Linear regression (the strongest cross-sectional model) underperforms persistence on all metrics, confirming that sequential context is necessary for gains beyond the heuristic. Training-Mean negative R^2 confirms it performs worse than a constant prediction.*

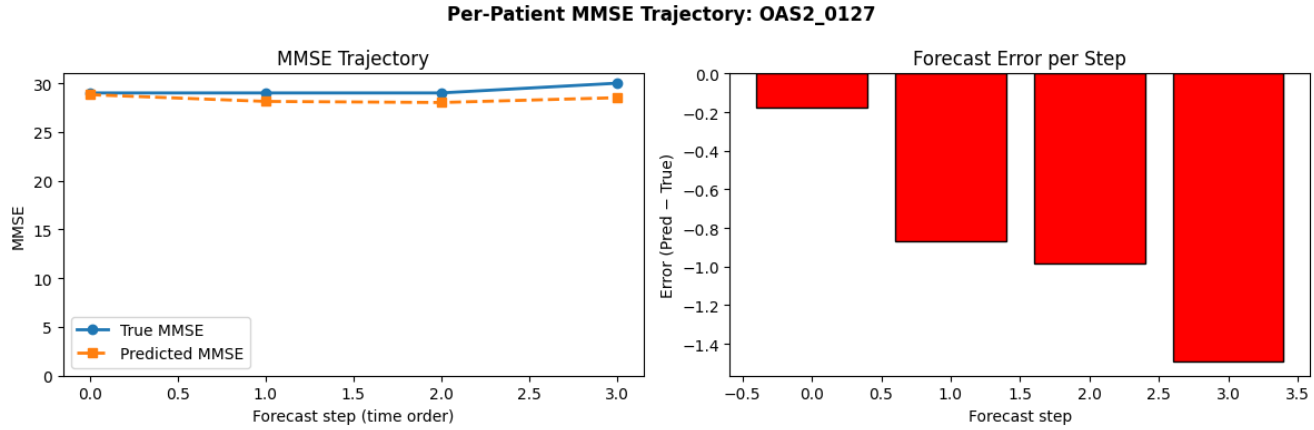

Figure 6. *Per-patient MMSE trajectory for subject OAS2_0127 (4 forecast steps — the test subject with most forecast points). Left: true MMSE (solid, circles) vs. predicted MMSE (dashed, squares). Both series stay near 30, with the model tracking the stable trajectory closely. Right: prediction error per step (Pred − True). All errors are negative and small (range −0.15 to −1.45), indicating a systematic slight underestimation for this high-scoring stable subject. This conservative bias is expected: the model is trained on the full population including declining patients and regresses slightly toward the dataset mean.*

5.5 Cross-Validation Results

Given the small dataset, a single 23-subject test evaluation carries high variance. Five-fold GroupKFold cross-validation was applied to the combined train+validation pool (127 subjects) using the best BiLSTM configuration. **A fresh StandardScaler was fitted within each fold's training partition** — never using validation data — maintaining leakage prevention throughout CV.

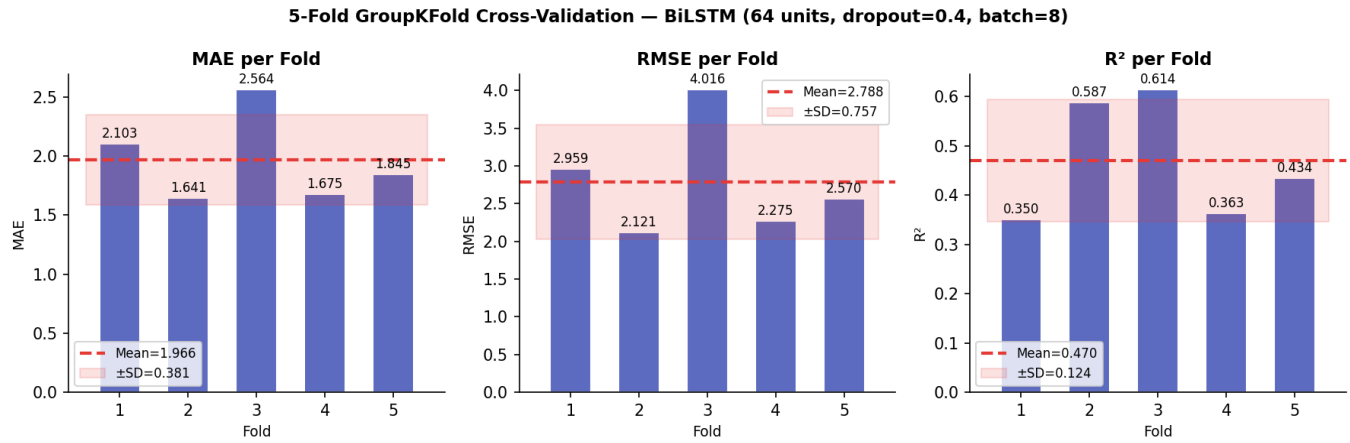

Figure 7. *5-Fold GroupKFold cross-validation results for BiLSTM (64 units, dropout=0.4, batch=8). Each bar shows the metric for one-fold; the red dashed line marks the mean; the shaded band shows ±1 SD. The widespread across folds — particularly Fold 3 (RMSE=4.02) — reflects the high sensitivity of performance to which 25 subjects land in each partition. Fold 3's high RMSE but elevated R^2=0.61 is characteristic of folds containing rapid-decline subjects: the model correctly tracks the direction of decline (high R^2) but with larger absolute errors (high RMSE). Mean R^2=0.47 ± 0.12 is the more conservative and reliable generalisation estimate.*

Table 6. *5-fold CV results with per-fold interpretation. The Mean row represents the primary performance estimate for this project.*

Fold	MAE	RMSE	R^2	Note
1	2.1026	2.9590	0.3504	Harder partition — mixed stable and declining subjects
2	1.6412	2.1208	0.5871	Favourable split; best overall fold performance
3	2.5639	4.0158	0.6143	Highest RMSE — dominated by rapid-decline subjects; high R^2 because model tracks the trend
4	1.6747	2.2749	0.3633	Moderate; lowest R^2 alongside Fold 1

5	1.8450	2.5697	0.4339	Near-mean across all metrics
Mean	**1.9655 ± 0.38**	**2.7880 ± 0.76**	**0.4698 ± 0.12**	**Conservative, robust performance estimate — preferred over single test-split result**

Reconciling test R^2=0.75 with CV R^2=0.47: The test partition happened to contain a higher proportion of stable, predictable subjects, yielding an optimistic result. With only 150 subjects, a 5-subject compositional shift between partitions can change R^2 by ~0.10. This is not a flaw in the model — it is a fundamental statistical property of small cohort evaluation. **We identify CV mean R^2=0.47 ± 0.12 as the primary performance estimate** and report the test result as an upper-bound reference.

6. Discussion

6.1 Why BiLSTM Outperforms Alternatives

At sequences of maximum length 4, the BiLSTM's bidirectional mechanism provides no long-range contextual benefit; the more parsimonious explanation is parameter count — a BiLSTM with 64 units has approximately twice the recurrent weight capacity of a unidirectional LSTM at 64 units (LSTM(64): ~21.8k trainable parameters; BiLSTM(64): ~43.6k parameters — computed as 4×(n_feat + units + 1)×units for each direction, plus the shared Dense layers), functioning as a larger model regularised aggressively via dropout (0.4) and small-batch gradient noise. Future work should compare parameter-matched architectures (e.g., LSTM with 128 units vs. BiLSTM with 64 units) to isolate the genuine contribution of bidirectionality.

6.2 Small Dataset and Persistence Baseline

The persistence baseline (R^2=0.72) is powerful because 36% of consecutive visit pairs show zero MMSE change (Figure 1, bottom-centre). The BiLSTM's 7% RMSE improvement (1.27 vs. 1.37) and R^2 gain (0.75 vs. 0.72) are driven by better handling of *changing* patients — the Demented and Converted subgroups. With only 150 subjects, a 5-subject partition shift changes test R^2 by ~0.10; CV mean R^2=0.47 ± 0.12 is the primary estimate. The cohort consists of right-handed adults aged 60–96 from a single institution; generalisation to broader populations requires further validation.

6.3 CDR in Clinical Deployment

CDR(t) is valid when the model is applied after a clinical visit. If used prospectively (forecasting before a visit), CDR would need to be excluded or separately estimated — an important distinction for any clinical deployment.

7. Conclusions and Future Work

7.1 Summary

This project delivered a formally defined, methodologically rigorous pipeline for next-visit MMSE forecasting on the OASIS-2 longitudinal cohort. Key contributions: formal sequence regression framing with mathematical notation; masking-safe per-timestep standardisation; patient-wise leakage prevention at every stage; 108-configuration sweep with validation-governed selection; and transparent dual evaluation (test + 5-fold CV with reconciliation). The BiLSTM achieves test R^2=0.75 and CV mean R^2=0.47 ± 0.12 — the latter being the primary, reliable performance estimate. All four learning outcomes of the course project brief are addressed: formal problem definition, justified technique selection, appropriate experimental evaluation, and complete implementation.

7.2 Future Work

- Larger dataset: integrating ADNI (Alzheimer's Disease Neuroimaging Initiative) would increase the cohort by an order of magnitude, substantially reducing all metric variances.
- Group-stratified evaluation: separate reporting per subgroup (Nondemented / Demented / Converted) would reveal where sequential modelling adds clinical value.
- Probabilistic forecasting: distributional outputs (quantile regression or MC-Dropout) would provide calibrated uncertainty intervals for clinical decision support.
- Attention and interpretability: self-attention over the visit history or SHAP analysis would quantify each feature's contribution to individual predictions, supporting clinician trust.

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